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# A review on current state of art of a blockchain interoperability

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ABSTRACT

Blockchain technology is one of the highly influential technology nowadays due to its features such as highly secure, immutable data storage. Many blockchain developments are in their infancy because blockchains work in their own silos and are not sufficiently interoperable with each other. Many studies are not sufficient to understand the process of working in interoperability of blockchain area. Therefore, this paper is to provide a sufficient survey on approaches of blockchain, transaction method and type of interoperability. This paper, a proposed flow of the interoperability process for any use case , where blockchain interoperability can be applied. This paper has a comprehensive study and observations for current state of interoperability of blockchains. This paper also provides different platforms which will help for inter operation between crosschain platforms.

## 1. Introduction

In this era of swift advancements in information technology,it is crucial for individuals and organizations to exchange information with one another. The development of blockchain technology is severely constrained by the isolation of blockchain networks, which makes it difficult for multiple blockchains to collaborate.

Blockchain interoperability referred as the capability to exchange digital assets and transaction information across different networks independently, without depending on or being limited by trusted third-party exchanges.

Blockchain interoperability [1] is still in the early stages of development. Currently, there are no widely accepted standards that enable seamless connections between different blockchain networks. This lack of standardization poses significant challenges [2] and highlights the need for extensive research and innovation. Efforts are actively being made to improve the ability of blockchains to interact and exchange information smoothly, allowing them to function together without the need for a trusted third party [3]. The goal is to create a unified ecosystem where blockchains can communicate effectively, unlocking new opportunities for decentralized applications and services.

### 1.1. Blockchain networks

Blockchain is a peer to peer, decentralized, immutable and distributed network. Satoshi Nakamoto first proposed blockchain in 2008 [4] . The modern era of blockchain technology places a premium on interoperability, making it imperative to focus on achieving flexibility in blockchain usage. An interoperable blockchain is made up of distinct blockchain systems, each of which represents a separate and unique distributed ledger . Data recorded in one blockchain can be verified, accessed, and referenced by another, and transactions between different blockchains or ledgers can also be made using this method.

### 1.2. Features of blockchains

*Immutable:* The blockchain's immutability indicates that it is an everlasting and unchangeable network. Blockchain technology is powered by a node collection. Once a transaction is added to the blockchain, it cannot be altered or deleted. This makes the blockchain a highly reliable and secure ledger that is tamper-proof and resistant to manipulation.

A digital ledger copy is maintained by each node in the network. Before a transaction is recorded, each node checks its validity, and it is accepted only if most nodes approve it. This ensures that no transaction block can be added without the majority's agreement. Once confirmed, the records become permanent and cannot be altered or reversed. This implies that it cannot be edited, changed, or deleted by any user on the network. The figure in 1 shows the features of blockchains in pictorial presentation.

*Decentralised:* Blockchain technology operates on a decentralized network, meaning that no single entity controls the whole system.

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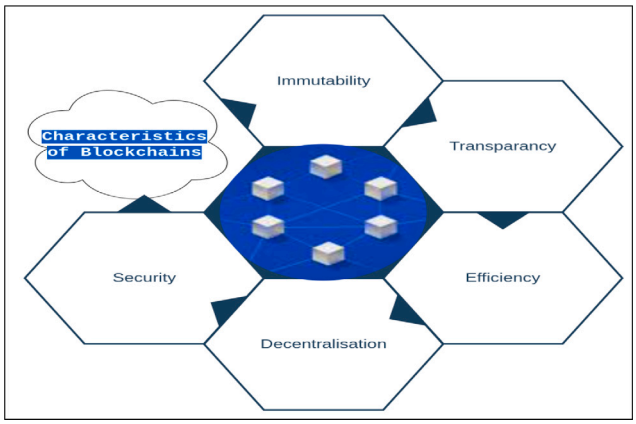


**Fig. 1.** Features of blockchain [6].

Instead, a large network of nodes works together to verify and approve transactions. As a result, an identical copy of the ledger hold by each node in the blockchain network.

Many benefits stem from the decentralization attribute in the blockchain network: A blockchain network remains highly organized and resilient because it does not depend on human calculations. The decentralized nature of the blockchain network makes it less prone to failure. Because it costs more money for the hackers to attack the system and less chances to be successful. The system is at low risk, as no third party is involved in it. Because blockchain technology is decentralized, it is easier to create an open profile for each member of the network. Every modification is therefore traceable and more creative [5]. Users no longer need to depend on outside parties to maintain and manage their assets; instead, they are in charge of own possessions.

*Secure:* Every entry in the blockchain has a unique encryption. Encryption enhances the overall security of the blockchain network. With no central authority, adding, modifying, or deleting data becomes extremely difficult. Because every piece of data is cryptographically hashed, it carries a unique identifier within the network. Each block contains its own unique hash along with the hash of the block that precedes it. Because of this feature, the blocks are connected cryptographically to each other. Any attempt to alter the data would changes each hash ID, which is not possible.

Anyone can access and view the network's transactions because the blockchain ledger is transparent and open to the public. As a result, it is an extremely transparent system that is impervious to corruption and fraud.

### 1.3. Types of blockchains

There are three main types of blockchains, that are public, private and consortium blockchains

#### 1.3.1. Public blockchains

The network is a public infrastructure, meaning that anyone can freely join without needing permission. Participants in the network have the ability to view, participate in, and validate transactions. Ethereum and Bitcoin are the most widely used infrastructures. Public blockchains (BCs) are primarily applications accessible to the public where complete transparency and decentralization are necessary. Furthermore, they incentivize network participants, known as "miners", to verify transactions by providing them with cryptocurrency rewards.

#### 1.3.2. Private blockchains

These networks enable invited participants to engage in specific transactions or additional block creation tasks. They are designed for organizations that desire restricted and private access to data, thereby keeping it inaccessible to the general public.

#### 1.3.3. Consortium blockchains

Multiple organizations can jointly manage a blockchain, with these selected entities deciding who is allowed to submit transactions or access information. Consortium blockchains are well-suited for businesses where all participants need authorization and share the responsibility of maintaining the network.

### 1.4. Interoperability

Within the cryptocurrency and bitcoin space, the process of enabling blockchain-related or blockchain-based technologies to communicate with another blockchains or other systems without blockchains is referred to as blockchain interoperability. Because blockchains are rigid by nature, no two blockchains can communicate with one other [7]. The majority of blockchains are thus incompatible with one another. But there are other blockchains out there, each with special characteristics of its own.

- Increasing compatibility of blockchains with each other and enabling their incorporation into current systems are the two main objectives of interoperability.
- The capacity of two businesses to collaborate on an existing system that only makes use of blockchain technology is known as interoperability. This allows both businesses to save money on blockchain management and compliance.
- The notion of interoperability facilitates the development of numerous novel apps built upon pre-existing frameworks, granting users access to data from various blockchains within a solitary application.

For example, Assume use of Ethereum as a base for smart contracts for an existing website, like Shopify or WordPress .

#### 1.4.1. Benefits of interoperability:

Interoperability has the benefits as follows:

- Flexible data exchange [1]: Currently, there is significant progress in blockchain applications, although they are primarily confined to single blockchains or operate within the isolated ecosystem of blockchains. Implementing a solution for blockchain interoperability would make data sharing and exchange more adaptable across different blockchains.
- Asset Exchange or Secured Data [7]: Assets may be traded concurrently across various blockchain ledgers; therefore, a secure interoperability channel is essential for data transfer or asset exchange.
- Ease of transaction with other blockchains [8]: Using interoperability, transaction can happen over different blockchains even if the blockchain networks have different architectures

#### 1.4.2. Definitions of interoperability:

Interoperability is a way of interconnection between two systems or networks for sharing data,assets transfer or exchange of information or currencies. Many author's and researcher's defined interoperability as follows: The National Institute of Standards and Technology (NIST) [9] states that data sharing is the main focus of blockchain interoperability, which is defined as "A combination of distinct blockchain systems, each functioning as an independent distributed ledger, where atomic transactions can be executed across multiple heterogeneous blockchains. In this setup, data stored on one blockchain can be accessed, verified, and referenced by transactions on another blockchain, provided there is semantic compatibility between them. . Pillai and Biswas [10] believed that "Cross-communication is not designed to directly alter the state of another blockchain network". According to the definition of blockchain interoperability. Rather, a set of functions on the other system that is anticipated to carry out an action within its own network should be

triggered by cross-communication. Ethereum founder Vitalik [11] mention that ”the capacity to move funds, assets, or information across two blockchains is the primary definition of “blockchain interoperability”. Besançon et al. [12], indicate that there are other kinds of blockchain-interoperable applications outside blockchain. Three different forms of interoperability exist for blockchains. Three types of interoperability exist: first, between different blockchains; second, between DApps (decentralized applications) operating on the same blockchain; and third, between blockchain technology and other technologies (such enterprise systems integration). Three categories for blockchain interoperability are suggested few research papers. The first is heterogeneous blockchain interoperability, which mostly refers to blockchains with various underlying architectures, such as encryption and consensus algorithms. Here the heterogenous blockchains are the chains which are working in different networks or are working in different platforms as well.

- Homogeneous Interoperability: Interaction between chains that are very similar (same VM, Similar Consensus, same address formats). Example: Ethereum Mainnet and ethereum sidechain or two EVM chains using same contract bytecode and accounts.
- Hereogeneous Interoperability: Interaction between chains that differ in VM, consensus, state model, or identity primitives (e.g. Ethereum and Bitcoin or Ethereum and Hyperledger Fabric). Requires adaption layers, canonical encodings, or richer middleware.

The structural interoperability of blockchain is involved in heterogeneous chain interoperability. Interoperability between isomorphic blockchains is the second. In reality, many consortium chains that are visible today have underlying architectures that resemble fabrics. Dapps can be defined as the interoperability of these blockchains with comparable underlying systems. The third is the capacity of blockchain technology to work with non-blockchain systems. For instance, the way the current financial system is integrated with the central bank’s sovereign public blockchain [13]. The mechanism of interoperability is involved in the interaction between non-blockchain and blockchain technology.

Mostafa Kazemi and et al. in their paper [14]defined blockchain interoperability as it is the idea of interaction and integration between multiple different blockchain systems.

As per our perspective, “Interoperability in between two blockchains is an interaction between two or more blockchains to transfer, share, exchange and distribute the data, facility or information for flexible, scalable and secure networks”.

### 1.5. Challenges in blockchain interoperability

As the blockchain networks are working in their own silos and are not interconnected. To interoperate blockchains there are some challenges to apply interoperability between different blockchains due to its different use, architectures, frameworks and permissions are different.

- Currencies of blockchains are different in case of blockchains made for cryptocurrencies. Example: Exchanges between Ethereum and Bitcoin
- Different smart contracts used in different blockchain networks, based on applications. Example: Ethereum uses python or solidity programming languages to write smart contracts however Solana uses either Rust as native language and sometimes uses C or C++ languages for smart contract writing.
- Separate blockchain networks with often different protocols and standards [6]. Examples: as bitcoin uses proof of work(PoW) consensus as of now and ethereum uses proof of stake(PoS).
- Operated in their isolation, from one another. For examples cosmos and polkadot are the platforms which can be operated for interoperability with their own closed environment.

From the above survey paper studies and from the research done till now about interoperability in blockchains, it is important to study and understand the current state of the cross-chain interoperability. Therefore, the following research requirements are decided for this paper.

- Research Requirements
  RQ1: Different Approaches for Blockchain Interoperability
  RQ2: Platforms available for Blockchain Interoperability
  RQ3: Issues in current interoperability of blockchains
- Research Conducted keywords used and search engine used for it are mentioned in 4
- Selection of Relevant Papers Inclusion and Exclusion criteria is also mentioned in 4 According to the research requirements this paper contributions as follows:

  1. This paper surveys systematically about different approaches in blockchain interoperability. Moreover, this surveys different popular platforms which offer blockchain interoperability related services.
  2. Different challenges, issues, advantages and disadvantages of different approaches are discussed.
  3. Current state of the research in interoperability of blockchains.
  4. It is the first survey to propose the workflow while working on interoperability case studies.

### 1.6. Road map of paper

The introductory information has been presented in Section 1, followed by the required background related literature study on blockchain in Section 2. The basic research methodology and survey method for the survey of the paper have been discussed and presented in Sections 3 and 4. Later, this paper have discussed various approaches for blockchain interoperability in Section 5, followed by the comparison among them. In Section 6 all the different blockchain and interoperability platforms are discussed in detail with the comparison of the same shown in table. Finally, this research have proposed a layered workflow for interoperability solution is provided in Section 8 and concluded in Section 9.

## 2. Related literature reviews

The Blockchain technology is growing gradually nowadays. This growth found an obstacle as interoperability in blockchains for further growth of blockchain technology. Many proposals are working in isolation, therefore, there is need to tackle this problem for further increase in flexibility of inter operation. Several literature reviews have been conducted to showcase proposed solutions addressing this issue, and this chapter outlines some of those surveys and compared for further need of survey in this area of research.

The paper [15] investigates the significance of smart contracts in addressing the challenge of blockchain interoperability. It provides a focused review on the role of smart contracts in achieving interoperability, offering a categorization of blockchain solutions that can interoperate based on blockchain types and the programming languages used for smart contract development. The paper also addresses various unresolved issues and identifies research opportunities for the future. Blockchain interoperability is defined as the ability of a distributed ledger to process transactions originating from another distributed ledger, encompassing considerations such as identity management,

cryptographic management [15], consensus mechanisms, and smart contract capabilities. The discussion further explores scalability in interoperable blockchains, distinguishing between vertical and horizontal scalability, and provides a detailed overview of each type.

The paper [16] conducts a survey of practical instances and compares state-of-the-art systems to highlight their unique features. It defines interoperability as the ability to transfer assets across a combination of homogeneous and heterogeneous blockchain systems without compromising each blockchain's original design philosophy. The paper categorizes research on blockchain interoperability into three major categories: chain-based, bridge-based, and DApp-based interoperability [16]. It provides a survey of available platforms within each category and discusses their advantages and disadvantages. Finally, the paper discusses critical challenges and suggests potential research directions for interoperability in blockchains.

In the paper [17], This paper offers a comprehensive survey of the existing cross-blockchain communication solutions, accompanied by a comparative analysis of their proposed architectures. The primary contribution lies in categorizing these solutions into four main types: sidechain-based methods, blockchain routers, smart contract-based approaches, and industrial implementations. Additionally, the paper discusses the current state of blockchain interoperability, identifies the key challenges faced in the field, and suggests possible future research directions. It is important to note that the study focuses exclusively on interoperability among homogeneous blockchains. .

The papers [5,18] provide a comprehensive review of blockchain interoperability solutions categorized into three main directions: Notary schemes, Hashed Time-Lock Contracts (HTLC), and Relays (including Relay Chains as a subcategory). It discusses the advantages and disadvantages of each solution, comparing them with each other and highlighting their target domains, distinguishing between arbitrary data exchange and digital value exchange. This paper's contributions include a systematic review of the state-of-the-art in heterogeneous blockchain interoperability, comparison of reviewed solutions on six different criteria, discussion of open challenges in blockchain interoperability, presentation of research directions, and the introduction of a taxonomy for classifying current solutions. Additionally, possible new research directions for incremental progress in existing solutions or develop new ones are suggested.

The authors of paper [1] defined interoperability layers as technical interoperability, semantic interoperability, organizational interoperability, legal interoperability, integrated public service governance, and interoperability governance. This paper focuses on two layers for interoperability as technical and semantic interoperability in the context of blockchain technology, noting its potential to address major research challenges. As a key research, the paper introduces the Blockchain Interoperability Framework (BIF), which assesses interoperability solutions based on various criteria such as Public Connectors, Blockchain of Blockchains, and Hybrid Connectors [19]. The review encompasses diverse sources, including peer-reviewed papers, white papers, and industry solutions, also provided comprehensive insights and future directions for research. Additionally, the paper identifies real-world use cases and challenges in developing interoperability solutions and proposes avenues for future research as this field is rapidly evolving. They also discussed different blockchain interoperability architectures and standards.

Monika and Rajesh Bhatia in their paper [20] mentioned interoperability solutions w.r.t its use cases as asset exchange, asset transfer and cross-chain oracles. They have mentioned approaches such as notary, sidechains/relays, smart contract solutions and bridging solutions, but the study is limited to only two years survey also nothing mentioned about research directions in future.

In paper [21], Reference solutions are provided, the challenges of cross-chain technology are examined, and the transaction, interaction, asset management, security, and other aspects of cross-chain technology are explored. They said that the process of continuously resolving current problems is what drives the cross-chain technologies development to obtain a more thorough grasp of cross-chain technology; the entire book is condensed, and the extensive research material and areas of cross-chain technology focus are projected until the year 2018.

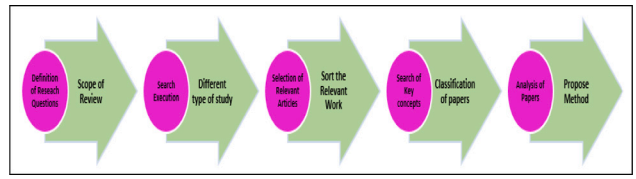


**Fig. 2.** FLow of research.

According to research paper [22] authored by HaonanYuan, and et al. there are some security issues in current interoperability solutions, which need to pay an attention to the privacy and security aspects of blockchain interoperability.

## 3. Survey methodology

### 3.1. Research objectives aligned with methodology

This study is guided by three main research questions:

- **RQ1:** What are the current approaches for achieving blockchain interoperability?
- **RQ2:** What platforms support interoperability between heterogeneous or homogeneous blockchains?
- **RQ3:** What are the prevailing issues and limitations in the implementation of blockchain interoperability?

These questions form the foundation of the methodology, guiding the search strategy, inclusion criteria, data extraction, and comparative analysis. The methodology has been structured to systematically address each of these questions through targeted literature review, screening, categorization, and synthesis.

The objective of this article is to evaluate the existing level of blockchain interoperability in the literature and to pinpoint any gaps that may be filled to forward the creation of interoperability solutions. However, because there are so many blockchain architects that have been proposed, there are a lot of interoperability issues in the real world because multiple blockchain architectures are being deployed. Indexing databases were searched in order to gather literature.

## 4. Research methodology for survey

The database searches utilizing the different search keys are displayed in Table 1. The figures in the table represent the initial search's unfiltered results. Table 2 displays the outcomes following. The figures in the table represent the initial search's unfiltered results. The outcomes of applying the different filters are displayed in Table 3. A flow of the research methodology process is presented in Fig. 2. A summary of the paper inclusion criteria is presented in Fig. 3. To further demonstrate the advancements being made in both the academic and industry domains, a few well-known blockchain interoperability solutions that have been suggested in the industry were also presented. The distribution of the papers included in this survey is displayed in Fig. 3. Additional resources include pre-prints, papers, and websites.

### 4.1. Limitations of the methodology

While the review process was systematic and structured, the following limitations are noted:

– The study relies on secondary literature and does not include implementation-based testing.

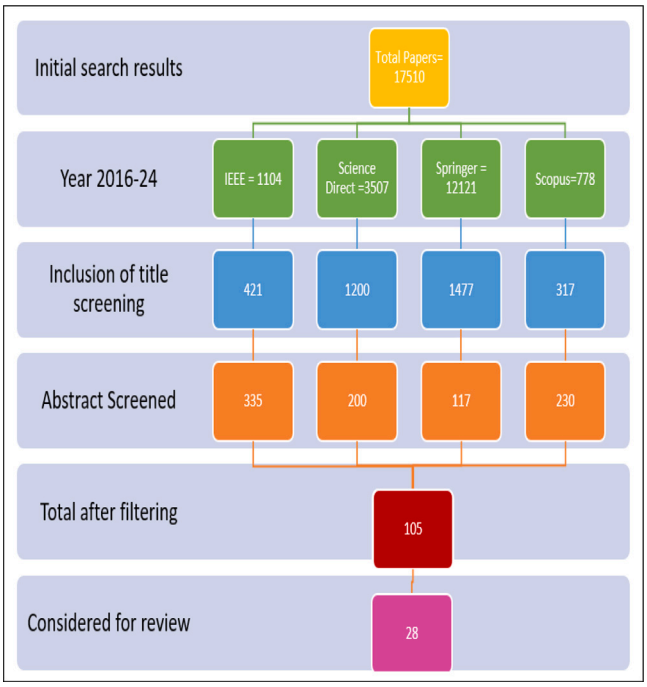


**Fig. 3.** Survey screening methodology.

– Comparison is based on qualitative data due to lack of standard interoperability benchmarks.
– Security and privacy implications were not deeply analyzed during classification.
– Workflow validation was conceptual; future work will focus on empirical testing.

### 4.2. Inclusion and exclusion criteria

The following inclusion and exclusion criteria were applied during literature selection to ensure relevance and quality:

**Inclusion Criteria:**

– Peer-reviewed journals and international conference papers published in between year 2020 and 2024.
– Studies that explicitly discussing blockchain interoperability approaches, frameworks, or platforms.
– Research providing experimental results, real-world implementations, or simulation-based evaluations.

**Exclusion Criteria:**

– Studies focused solely on blockchain security without interoperability relevance.
– Articles without technical contributions, such as opinion pieces or conceptual editorials.
– Duplicates or papers with insufficient methodological detail.

### 4.3. Search strategy and databases

The literature search was conducted between January 2016 and April 2024 across the following digital libraries:

– ScienceDirect
– ACM Digital Library
– IEEE Xplore
– SpringerLink
– Google Scholar

**Search strings** included combinations such as:

– "blockchain interoperability"
– "cross-chain platforms"
– "blockchain data exchange"
– "multi-chain transaction" AND "performance"

Boolean operators (AND, OR, NOT) and filters for years (2020–2024) were used. Manual screening of abstracts and full texts was done to apply the inclusion/exclusion criteria.

### 4.4. Data extraction and analysis

Key data from each selected study were extracted manually into an Excel spreadsheet and included:

– Approach or architecture used (e.g., notary, relay, sidechain).
– Platform(s) evaluated or proposed.
– Use cases or application domains.
– Performance indicators: scalability, security, decentralization.

Interoperability solutions were categorized manually based on architectural traits and validated through cross-reference with existing surveys. Comparative analysis used metrics derived from published data where available.

## 5. Approaches for blockchain interoperability

From the literature review, there are many approaches to achieve interoperability. We differentiate approaches as notary scheme, bridges and relays, hash lock time lock, sidechain, blockchain engine, gateway based and other approaches also studied in this paper. Defined categorization are shown in Fig. 3.

S. Zhu and et al. in their paper [13] mentioned that for a successful blockchain interoperability solution, it needs to solve five problems. (a) Consistency and Atomicity of cross-chain transactions; (b)cross-chain identity management; (c)cross-chain security; (d)cross-chain communication protocol; (e) crosschain trust mechanism.

Here in this paper, by observing the overall literature, approaches are divided in five types as Notary schemes, Bridges/Relays, Sidechains, Hash lock time lock and Blockchain Engines. These are explained in detail as follows:

(1) **Notary Scheme:**
Notary scheme is defined as a trusted individual or a group of the trusted third party is involved for blockchain interoperation [1,23]. Notary helps to validate data of one blockchain to be a part of another blockchain. It uses subset of the trustworthy servers [16,24] The servers need to prove the ownership and existence of the given asset within a delta time. The timed proof of existence feature of notary used for checking ownership to retain the immutability feature of blockchain. Notary scheme is one of the easiest way of interoperation with more centralization of the network [11].
**Advantages and Disadvantages** two major advantages of notary scheme are as follows: it is simple, because no need of extra changes are recommended in the original blockchains. Another benefit is trustworthiness distributed to the notaries. They use consensus protocols, for example, Byzantine Fault Tolerant (BFT) protocols [25] to reach agreement and provided the outcome as a trusted agreement. Also the agreement gets confirmed and considered valid if two-thirds of all the notaries set [24] have same agreement. But the disadvantage for notary schemes is it utilize the trusted third party as an intermediary between blockchains, and thus compromises the decentralization.
**Platforms use Notaries**. Herdius [1] is a decentralized exchange platforms use the notary scheme, as a common intermediary between blockchains. Bifrost is another project which uses the notary based solution, to interact with multiple blockchains [16].

(2) **Bridges/Relays:**
Bridge-based or Relays based interoperability targets the implementation of a component called as bridge, across blockchains for connection [26]. Compare to notary schemes, relays are used by relay schemes to achieve blockchain interoperability . Relays are system components that read and can validate the state of another blockchain, Different types of relays are one-way and two-way. One-way relays enable a blockchain to access or read the state of another blockchain; however, the opposite direction of communication is not supported. For instance, BTCRelay [27] verifies and validates transactions of Bitcoin, within the Ethereum ledger [28]. In contrast, two-way relays enable both blockchains to read and verify each other's states. . Dogetherum [29]is a two-way relay scheme that enables the transfer of DOGE tokens from Dogecoin to Ethereum and back from Ethereum to Dogecoin.
**Advantages and Disadvantages:** Bridges connect chains with different architectures and support token transfers, message passing, events, therefore these are flexible in nature. Once a bridge exposes messages on both sides, smart contracts can use them to build complex cross-chain flows. Some designs use light clients or trusted relayers to improve latency. Thus these are some advantages of bridges/ relays. Some disadvantages in these are, If temporary conflicts not solved,inconsistencies may occur, therefore dual states can be notes. As an example, Ethereum needs the capability to accept and verify messages that represent asset transfer events originating from the Dogecoin network. .

(3) **Sidechains:**
Sidechains [16] are those which have abilities to follow mainchain. Sidechains uses different consensus algorithms that will be of its pegged mainchains. Meanwhile, sidechain uses two-way peg and connected to its main blockchain as a secondary blockchain. Any two-way peg's central component is a relay routine that moves consensus and data between blockchains [16].
**Advantages and disadvantages**
Sidechains will start working for interoperability if and only if both parties are ready. But the Sidechain solutions are still in their early stages and have a number of unresolved problem, such as Centralization, Complexity, Instability
**Platforms of Sidechain** Loom, RootStock (RSK), Liquid, and Proof-of-Authority (PoA) networks are some examples using sidechain for interoperability within blockchains.

(4) **Hash lock Time lock:**
Hash-locking is the transfer of assets without the use of a reliable intermediary. By locking transaction using time lock and a hash used for transaction is the hash-locking approach. This works as atomic swaps like, it only keep the transaction locked,if both the parties agree on its obligations. Through the hash operation [16] and time difference exchange is achieved.
**Advantages and Disadvantages.** The hash-locking method is not a stand-alone fix. It still has a lot of disadvantages because a hash-locking solution needs to lock assets for an established transaction channel during the opening phase, which puts the user at danger of losing assets in the event of a timeout. This can result in a status that is inconsistent. The usual communication route can be blocked by malicious participants by issuing numerous fraudulent transactions, which will have a major impact on other honest participants' legal transactions.
**Platforms of Hash-locking:** CheaPay is a network of payment channel, that offers an off-chain transaction settlements between the blockchains. An HTLC technique is used by a PCN network to ensure atomic asset switching.
In a similar vein, CHTLC (Chameleon Hash Time-Lock Contract) is another tool that offers PCN networks privacy-preserving support during the asset transfer procedure. Atomic swaps based on HTLCs are made possible by the protocol stack known as Commit. It offers two different kinds of protocols:communication protocols and cryptographic protocols. With Atomic Loans, you can move money between different cryptocurrency systems without going via a third party. The loan process is divided into four stages: the load period, the bidding period, the seizing period, and the return period. It employs an atomic swapping mechanism that is based on an HTLC. A technological prototype of an atomic transfer procedure with a hash lock is offered by AltChain. The prototype is also the first utilized by the lighting network [16] to guarantee the atomicity of cross-chain transactions and perform contract locking and unlocked execution as part of the BTC off-chain transfer expansion solution [16].

(5) **Blockchain Engines:**
blockchain engine needs an infrastructure which is shared, for supporting services on different layers of blockchain, including consensus, network, incentive, etc.
**Platforms of Blockchain Engines:** Cosmos [30], Polkadot [31], Wanchain [16], ARK [16] are some platforms which are the industrial solutions
**Advantages and Disadvantages:** It is one of the scalable solutions, a multi chain interoperability also possible, but these are platform specific solutions and need some changes in original blockchains. As it require changes it is one of the complex solutions.

(6) **Gateway based:** The gateway such as trusted anchors acts as a middle ware between the permissioned blockchains and used to communicate as well as translates the messages received from the source blockchain to the target blockchain [32]
**Advantages and Disadvantages** It is an approach Without third party intermediaries, scalability can be achieved. Even though it has more advantages as fast and it is simple etc.

(7) **Other Approaches:**

- Smart contract based: In this classification of interoperable blockchain solutions is provided on the basis of blockchain type and programming languages used to develop smart contracts used in the interoperable blockchain solutions [15]
- Federated Learning and Dapp based Interoperability [37] A framework for federated learning utilizing a Directed Acyclic Graph (DAG) is proposed, which employs a sharding mechanism to improve blockchain interoperability. Sharding is nothing but it efficiently partitions the dataset while ensuring the requirements of federated learning are met.

Refer figure (see Fig. 4) and a comparison table for all approaches is shown in table no. 1

## 6. Blockchain platforms for different purposes

Many different developing platforms are exists for blockchain application developments. Each one of them has some of its features are unique. These are explained in this section in detail.

### 6.1. SOLANA

Solana is a permissionless blockchain platform, its more helpful in building cross-industry solutions, consensus algorithm used by Solana is Practical Byzantine Fault Tolerance (also known as Tower BFT)

The Solana [38] blockchain development architecture is designed to be ultra-quick, extremely safe, and censorship-resistant all at the same time. It is now one of the open-source, programmable Blockchain systems that is experiencing the most rapid expansion. When it comes to the development of decentralized apps (DApps) and marketplaces,

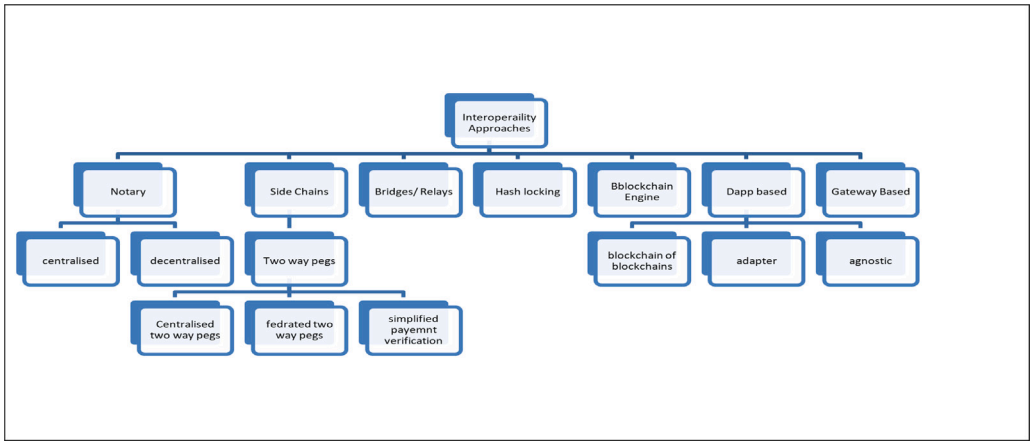


**Fig. 4.** Blockchain interoperability approaches.

**Table 1**
Different interoperability approaches.

| Approach | Properties/Definition | Platforms/ Projects | Advantages | Disadvantages |
|---|---|---|---|---|
| Notary [16] | A notary is a reliable person or group of people who are able to oversee several chains. | Herdius, Bifrost [1] | Timed proof of existence, Relatively simple, No additional changes are required in the underlying blockchains, distribute trustworthiness to the notaries | Not fully decentralized, mostly used for exchanges and not for data transfer, the third entity is utilized as trusted intermediary between blockchains |
| Side chains [26] | Sidechains are the other supporting chains added in parallel to mainchains | Loom, Rootstock, Liquid,PoA Networks | Only after agreement by both involved parties upon obligations. will not directly affect main chain environment | Centralization, more Complexity, Instability. Sidechains may operate using consensus protocols that are entirely independent from those of its pegged mainchains |
| Hash locking Time locking [27] | A hash-locking solution its opening phase must lock assets for a transaction channel that is established | CheaPay, CHTLC (Chameleon Hash Time-Lock Contract)[16], Commit, AtomicLoans, AltChain, the lighting network in the BTC used it first [16] | It is a solution without intermediaries, More decentralization | The user bears the risk of losing their assets in the event of a timeout. An inconsistent status could result from this. Only asset exchange work available |
| Bridges/ Relays | Use relays to achieve decentralized blockchain interoperability. | BTCRelay [16], Dogetherum | Employing relays as system components capable of verifying and accessing the state of a different blockchain. | Decentralization is compromised. Relays can be malicious |
| Blockchain Engine [33,34] | Parachains are used, For blockchain engines to support services on several layers [16], such as incentive, consensus, network etc., a shared infrastructure is usually necessary. | Cosmos [34], Polkadot [31,35], Wanchain [16,18], ARK [18] | scalable solution, Multi chain interoperability is possible | Platform specific, need some changes in original blockchains, complex solution |
| Gateway Based [32] | Trusted anchors are used for communicate between blockchains | Ethereum and Hyperledger [36] | Without intermediary, scalability can be build, | Consensus on consensus mechanism need to improved |

developers and organizations from all over the world rely heavily on a Blockchain framework known as Solana. This is partly attributable to the fact that it had special properties, one of which was the broad acceptance of it as a Blockchain. This adoption made it possible to develop open infrastructure in a flexible manner.

One of the open-source programmable Blockchain platforms with the quickest growth, Solana is a lightning-fast, safe, and censorship-resistant Blockchain architecture. Solana's unique qualities led to its mass adoption as a Blockchain framework. It is extensively utilized by developers and organizations worldwide to create decentralized apps (DApps) and marketplaces, and it aids in the flexible construction of open infrastructure. Going for Solana development can make you very competitive in terms of speed and scalability compared to other available Blockchain platforms. Solana is one of the blockchain frameworks that can process more than 1000 transactions per second, making it the most scalable and the first genuinely web-scale blockchain. As a Blockchain platform, Solana has its native token in the name of SOL that was traded at 65.06 US dollars having a 286 million token circulation supply and a maximum supply of 489 million coins. The day-to-day affairs of Solana are looked at and managed by the Solana

Foundation, which is a non-profit organization based in Switzerland, and the advancement in terms of technology development is done by Solana Labs.

Features that make Solana unique among other Blockchain platforms Solana is a ultra fast in speed of transaction . It is one of the scalable and highly secure solution. It is suitable for DApps and NFT marketplace solutions.

### 6.2. CARDANO

CARDANO is a permissioned blockchain platform. It focuses on Cross industry functionalities. CARDANO used Proof of stake as consensus algorithm to reach an agreement.

Cardano [39] is the blockchain platform with the highest rate of growth. It is built on the Proof-of-Stake (PoS) system, which was developed through peer-reviewed research. In addition to co-founding Ethereum, Charles Hoskinson began working on Cardano in 2015. Cardano is now Ethereum's rival, promising scalability, interoperability, and sustainability on its network.

The Cardano Computational Layer (CCL) and the Cardano Settlement Layer (CSL) comprise the two tiers of the Cardano smart contract system.

As per sources, a new upgrade is about to arrive named Alonzo and it is anticipated that with it there will be several improvements to this Blockchain platform regarding the smart contract functionality. Ada is the native currency of cardano, which is named after a 19th-century countess and English mathematician called Ada Lovelace, who is known as first computer programmer. The Cardano Foundation is the organization that looks after the promotion and standardization of the Cardano Blockchain platform.

Features that make Cardano unique among other Blockchain platforms are, It is a two tired Architecture, highly scalable and needs low gas fee for transactions.

### 6.3. TRON

Tron is a permissioned blockchain. It also helps to solve cross industry problems through blockchain. Tron uses proof of stake consensus.

Tron [40] is a high-performing decentralized Blockchain network that strives to develop a decentralized web. Tron functions over the Proof-of-Stake (PoS) mechanism algorithm to reach consensus and its own mainnet which was launched in 2018 [40]. Tron is mainly developed to support applications of entertainment that leverage Blockchain technology and peer-to-peer networks to remove intermediaries. Tron was developed as a direct competitor to the traditional media industry with its aim to eliminate the middlemen from the process of consuming media by users. Tron has its own native cryptocurrency for the system, known as Tronix (TRX) and currently, it commands 31st position in terms of market capitalization ranking among cryptocurrencies. Features that make Tron unique among other Blockchain platforms are, It is also one of the scalable solutions with high throughput and low gas fee requirement. It is Ethereum Virtual Machine (EVM) compatible

### 6.4. KLAYTN

Klaytn is a permissioned ledger, it provides service-centric industrial solutions using Istanbul Byzantine Fault Tolerance (IBFT) consensus algorithm.

Klaytn [41] is an open-source service-oriented Blockchain protocol developed by Ground X, the South Korean social media company which traces its origin to Kakao. KaKao is a renowned publicly traded Internet company headquartered in South Korea. Klaytn was designed with a strong focus on enterprise needs, providing numerous features commonly found in protocols of public blockchains. It offers distributed governance, decentralized data management, high scalability and low latency making it well-suited for business applications. Klaytn is an Ethereum Virtual Machine (EVM) compatible Blockchain network and its native token goes by the name of KLAY, which is used to perform all transactions on the network.

Features that make Klaytn unique among other Blockchain platforms It is an enterprise friendly platform with high throughput speed and Ethereum Virtual Machine (EVM) compatible

### 6.5. ALGORAND

ALgorand is an industrial permissioned ledger platform, focuses on service centric over industry. It uses Pure Proof-of-Stake (PPoS) consensus algorithm

Algorand [42] is an one of the open-source Blockchain protocols which is making itself into the list of top trending platforms due to its great features. Algorand was ideated and created to solve scalability problems users face in the legacy of Blockchains like low throughput and high gas fees. Algorand works to reduce the amount of energy consumption to do computation, and functions in an efficient way to create a distributed ledger with a unified system, without being forked or splitting of Blockchains.

Algorand comes with its unique combination of open-source Layer-1 capabilities, that opens up a world of new possibilities for developers. Currently, there are around 100,000 developers doing Algorand development as summarized by Algorand's CEO Steve Kokinos. Algorand developers are in great demand due to the innate qualities of Algorand of automatically enforcing custom rules and logic that have been provided in Algorand's ASC1s so that Algorand becomes cost-effective and error-free.

Features that make Algorand unique among other Blockchain platforms

It has two tired Architecture. Scalable with low gas fee requirement. It is Eco-friendly by consuming less energy.

### 6.6. POLYGON

Polygon is also a permissioned ledger. It uses Proof-of-Stake Plasma based-sidechain (PoS) consensus algorithm.

Polygon [43] is an Ethereum Virtual Machine (EVM) compatible Blockchain network created to solve the long-pending problems of scalability, and usability. Polygon was inspired due to the increasing demand for decentralized financial (DeFi) services coupled with the unpreparedness of the existing blockchain ecosystems to cope up to the growing needs of the time. Polygon's token Matic is the native token of the Polygon network that is a popular token among cryptocurrency enthusiasts, investors, and companies hoping to benefit from innovations in Blockchain technology.

Golang is an open-source programming language used primarily for Polygon development to make an extensible modular framework which builds a Blockchain network compatible with an Ethereum. One of the recently released features by the Polygon is the Polygon Edge. Polygon Edge is a new modular framework that is to be used extensibly by developers for building Ethereum-compatible blockchain networks, sidechains, and general scaling solutions.

Features that make Polygon unique among other Blockchain platforms It has multichain architecture. it provides a high speed and scalable solutions with Ethereum Virtual Machine (EVM) compatibility.

### 6.7. FANTOM

It supports both ledger type as permissioned and permissionless and uses a different consensus algorithm as Lachesis.

Fantom [44] is an open-source, decentralized, permissionless, smart contract platform built especially for building decentralized applications (DApps) and securing digital assets. Fantom strives to provide more scalability with lower costs than those the legacy first-mover smart contract platforms offer. Fantom comes with a unique feature

in offering to users, where users are free, rather than depending entirely on Fantom's primary consensus layer, they build and implement their own separate networks. The primary motive for introducing this feature is to avoid network congestion and provide greater scalability to the whole ecosystem. FTM is the native token of the Fantom blockchain ecosystem, serving key roles such as enabling staking, facilitating governance, and supporting payments across the network.

Features that make Fantom unique among other Blockchain platforms It supports speedy transactions and scalable solutions with low gas fee and Ethereum Virtual Machine (EVM) compatibility.

### 6.8. Near protocol

It is used as a permissioned ledger in blockchain platforms. It supports Proof of stake consensus algorithm.

Near protocol was initially developed as a machine learning protocol but its creators realized that no other existing Blockchain protocol can match up to their needs and aspirations. During the initial stages of the investigation, the creators Illia Polosukhin and Alexander Skidanov learned about smart contracts and cryptocurrency, which made them learn more about cryptography and conduct more studies.

Near Protocol operates on a Proof-of-Stake (PoS) consensus mechanism and was officially launched in 2020. Within the first four months, it secured $50 million in funding. The protocol is developed by Near Collective, the parent organization, which currently has a team of over 80 developers. .

Features that make Near Protocol unique among other Blockchain platforms Here transaction speed is fast, allows token transfer from one another blockchains. It is more suitable solution for minting NFTs.

### 6.9. CORDA

CORDA is a permissioned ledger. It uses Asynchronous Byzantine Fault Tolerance consensus algorithm

Consortium-based Corda [45] is the first of its kind of open-source Blockchain-inspired DLT (Distributed Ledger Technology) platform built exclusively for businesses. By embracing Corda, you can join numerous developers building applications over Corda, which is known as CorDapps. Over 60 companies from various industries have been working with enterprise software firms to R3 to develop distributed ledger technology (DLT) based on various applications on the Corda platform. Which aims to remove costly friction in business transactions along with interoperability and a unique approach to secure data privacy. Corda is a permissioned blockchain framework that only allows authorized participants to access the data, also it does not have any inbuilt token mechanism nor does it have any cryptocurrency. Owing to its fine-tuned access control to digital records Corda was initially designed for the financial industry but now it is a top Blockchain platform to be widely in use over sectors such as supply chain [46], healthcare, and government authorities. Several firms such as Bank of America Merrill Lynch, HSBC, and Intel along with others have invested a large sum into R3 Corda.

Features that make Corda unique among other Blockchain platforms It is more used for easy integration for enterprises, with high scalability.

### 6.10. ETHEREUM

Ethereum is a permissioned cross industrial solution, uses proof of stake as a consensus algorithm.

Ethereum [47]is an open-source blockchain-based distributed computing platform and operating system having smart contracts functionality. The cryptocurrency researcher and programmer Vitalik Buterin is responsible for Ethereum's birth. It was his idea that brought Ethereum to life in late 2013. Later on, 30th July 2015 when its system went live, it had grown manifold. Unlike other Blockchain frameworks, in Ethereum, anyone can do programming, and while talking about developers, they can use it to develop many new kinds of applications. Ethereum is a global, open-source Blockchain framework for decentralized applications and it was the first Blockchain framework to include a programming language. Ether is the cryptocurrency of Ethereum, which is much similar to Bitcoin and Ethereum allows one to send and receive coins between wallets. Ethereum-based DApps now ensure permanency, transparency and complete security to all electronic business transactions, due to recent advancements over time. In case your business is scalable, Ethereum-based apps allow you to leverage numerous advantages from Ethereum blockchain development to help increase your reach exponentially.

The Enterprise Ethereum Alliance now consists of more than 250 members, which consists of leading corporations like J.P. Morgan Chase, Microsoft, and Intel. Also, it connects nearly 500 Fortune enterprises, academics, start-ups, and various technology vendors at present. Many organizations and projects that build on Ethereum, such as decentralized finance (DeFi) platforms, prediction markets, and supply chain solutions, are likely to be interested in hiring Solidity developers. Other blockchain platforms that support smart contracts and may be looking for Solidity developers include EOS, TRON, and Tezos. It is worth noting that some of these platforms have their own native programming languages.

Features that make Ethereum unique among other Blockchain platforms Ethereum supports easy deployments . It has revolutionary Ethereum Virtual Machine (EVM) functionality.

### 6.11. TEZOS

It is a permissionless blockchain ledger. It uses delegated Proof of Stake consensus algorithm.

Tezos as a Platform whose co-founders are Arthur Breitman and Kathleen Breitman, which was built by them since 2014 with developers' team. In 2016 Breitman received around 612,000 dollars within 6 months by conducting an ICQ. Tezos is a decentralized blockchain, self-governed, platform that established a true digital commonwealth.

When compared to other Blockchain systems, Tezos is incredibly affordable because its Delegated Proof-of-Stake (PoS) consensus process requires just the participants to supply the minimal computational power to maintain network functionality. As a blockchain platform, it has a digitized token, called as Tez or Tezzie , where token holders rewarded for participation in the proof-of-stake consensus mechanism, a platform is independent of Tez mining process. Just like Ethereum, Tezos is a platform known for smart contract and dApp, but what makes it into a Top Blockchain platform is its self-amending cryptographic mechanism.

Features that make Tezos unique among other Blockchain platforms It is on of the on-chain Governance provider. It uses Self-Amendment Protocol and Functions over efficient Delegated Proof-of-Stake [17] consensus mechanism.

### 6.12. Hyperledger frameworks

Hyperledger is a big name when we talk about Blockchain development, as they have a variety of frameworks catering to the needs of different enterprises. You need not go elsewhere to look for an alternate Blockchain having varied tools and frameworks. Here are those frameworks developed by Hyperledger development, which are designed to address specific needs of enterprises.

### 6.13. Hyperledger fabric

Hyperledger fabric is a permissioned blockchain uses pluggable consensus framework. Hyperledger Fabric is a project of Hyperledger which is thought to be a base for developing solutions or applications with a modular architecture. What distinguishes Hyperledger Fabric is its modularity in architecture which allows in the plug-in for their

preferable components such as consensus algorithms and subscription services. Hyperledger Fabric with its modular and flexible design satisfies a wide selection of business use cases, and it gives an exceptional way of building consensus that permits functionality at scale while maintaining privacy. Therefore, many leading blockchain companies prefer Hyperledger Fabric as a Top Blockchain platform for building enterprise-grade applications.

Features that make Hyperledger Fabric unique among other blockchains are, it provides a high modular framework with Solidity and Ethereum Virtual Machine (EVM) support and a pluggable consensus mechanism.

### 6.14. Hyperledger Sawtooth

The Linux Foundation introduced Hyperledger Sawtooth, an enterprise-grade, modular Blockchain technology with contributions from IBM and Digital Asset. By dividing the core system into the application domain and providing a modular and flexible architecture, Hyperledger Sawtooth allows smart contracts to specify software's business rules. Hyperledger Sawtooth supports consensus techniques such as Proof of Elapsed Time (PoET) and Practical Byzantine Fault Tolerance.

Hyperledger Sawtooth is the second Blockchain framework that transformed itself into a Top Blockchain platform to be developed by Hyperledger and is an integration with an enterprise-ready 1.0 version.

Features that make Hyperledger Sawtooth unique among other Blockchain platforms are, It provides ability to do parallel transaction. Compatible with Ethereum.

### 6.15. Hyperledger Iroha

Hyperledger Iroha is a framework of blockchain, hosted by the Linux Foundation for building secure, trusted, and fast decentralized apps. The hard work by core teams of Hyperledger was helpful, ensuring a consistent balance between openness in the development process and the achievement of key milestones Therefore, transactions in Fabric can be hidden, but the development process itself cannot be concealed.

Hyperledger Iroha is a straightforward and modular distributed ledger framework built on a consensus algorithm called Another Consensus, which is fast and highly secure, and also protects Iroha networks against failures or attacks from adversarial nodes. This Blockchain framework is extremely suitable for distribution chain and IoT usage cases, as Hyperledger Iroha is highly portable and it supports macOS.

Features that make Hyperledger Iroha unique among other Blockchain platforms are It has modular architecture and provides transparent process and smart contracts.

### 6.16. Hedera Hashgraph

It is a permissioned blockchain ledger enables smart contract functionality. It uses Asynchronous Byzantine Fault Tolerance

Hedera Hashgraph is a DLT-based solution, similar to blockchain frameworks, that operates on a graph-like structure in which all nodes exchange information with one another and report that exchange by creating a graph of connections. Where such data or information is kept in events. Hedera Hashgraph can authenticate over a million signatures per second and process hundreds of thousands transactions each second, and this makes Hedera Hashgraph development extremely fast. To ensure fairness the Hedera Hashgraph platform never allows a single user to prevent the flow of transactions into the community . Thus, it guarantees the consensus-based sequence of transactions, reflecting the order in which the community receives them.

Hedera Hashgraph functions over an asynchronous Byzantine Fault Tolerance consensus algorithm that is capable of securing the platform against potential attacks. The governance of Headers Hashgraph is taken care of by The Hedera Hashgraph Council, which states clearly in its terms that Hedera's governance guarantees that no group or individual may exert undue influence on the body as a whole.

### 6.17. RIPPLE

Ripple is a non-profitable Blockchain framework discovered in 2012, which has aim to connect digital asset exchanges, payment providers, banks and corporates through RippleNet minus any chargeback as a Blockchain network. Validating servers secure Ripple with an internal ledger that guarantees transactions based on consensus. Ripple currently serves over 100 customers, with nearly 75 clients at various phases of commercial implementation. These include initiatives like reducing liquidity costs through xRapid, facilitating payments across diverse networks with xVia, and offering cross-border payment solutions via xCurrent.

Features that make Ripple unique among other Blockchain platforms are such as it is ultra fast, highly scalable, secure and supports a wide range of NFT marketplaces.

### 6.18. QUORUM

Quorum is permissioned ledger, Voting algorithms used consensus in Quorum.

Quorum is an enterprise-focused top blockchain framework that is based on Ethereum, founded by J.P. Morgan. It is an open-source Blockchain framework that is forever free to use. It is permissioned, which used different algorithms and vote-based algorithms to process hundreds transactions per second. It handles a higher throughput processing required applications with speed in transactions, which are private. Quorum addressed the confidentiality challenges that Ethereum and other blockchain platforms struggled with by implementing both private and public on-chain transaction options .

Features that make Quorum unique among other Blockchain platforms are, It is ultra -fast, scalable, secure and it supports wide range of marketplaces.

### 6.19. EOS

EOS is a permissioned blockchain ledger. Consensus used by EOS is Delegated Proof of Stake algorithm.

EOS is one of the blockchain platforms, a private entity named Block has founded it. One, as an open-source Blockchain framework in June 2018. The goal of the EOS platform is to solve the scalability problems of Ethereum and Bitcoin by providing decentralized storage of enterprise solutions, decentralized application hosting and smart contract functionality. There are no costs for users who utilize DApps based on EOS. EOS achieves consensus through the use of delegated proof-of-stake algorithm and multi-threading. Investors can discuss the platform on their own community forum, called EOS Forum.

### 6.20. DRAGONCHAIN

This is a permissioned, smart contract enabled platform. Consensus algorithm used by DRAGONCHAIN is Asynchronous Byzantine Fault Tolerance.

The Blockchain as a Service Platform from Dragonchain is both public and private known as a hybrid blockchain system and it was created at the Walt Disney Company. It was made available to the public later in 2016 with the goal of giving businesses and developers the tools they need to quickly create blockchain applications. Dragonchain gives businesses flexibility by permitting the use of Interchain. Their cloud-based architecture also eliminates the need for server setup, upgrades, and patch installation. It supports many programming languages, including Node.JS, Python, C#, Go, and Java, making it a Top Blockchain platform. Dragonchain provides five levels of consensus, allowing for a broad range of trust. It also Supports a wide range of NFT marketplaces

**Table 2**
Different blockchain platforms.

| Sr. no. | Platform/ Protocol name | Ledger type | Language support by platform | Smart contract enabled | Consensus algorithm | Governance | Token name | Features |
|---|---|---|---|---|---|---|---|---|
| 1 | Hyperledger fabric | permissioned | Python, GO, Java Script | yes | Pluggable Framework | Linux Foundation | No | Low latency Highly Modular Framework Solidity and Ethereum Virtual Machine support Pluggable Consensus Mechanism |
| 2 | Solana | Permission-less | Rust , C , C++ | yes | PBFT | Solana | SOL | Ultra-Fast transaction speeds Highly Scalable Highly Secure Suitable for development of DApps & NFT marketplaces |
| 3 | Ethereum | Permissioned | Golang, solidity | yes | PoS | Ethereum Developers | ETH | Easy Deployment Great Privacy Smart contract functionality available offers revolutionary ethereum Virtual Machine (EVM) functionality |
| 4 | polygon | Permissioned | Go lang, solidity, Vyper | yes | Proof of stake plasma based sidechain | Polygon Foundation | Matic | Multi-chain Architecture High speed & Highly Scalable Low gas Fee Ethereum Virtual Machine (EVM) Compatible |
| 5 | Cordano | Permissioned | Haskell as native | yes | POS | Cardano foundation | Ada currency | Two tired Architecture Highly Scalable Low Gas Fee Highly Eco-friendly by consuming less energy |
| 6 | TRON | Permissioned | Python, Solidity | yes | POS | the Tron foundation | Tronix (TRX) currency | Great Scalability High Throughput Low Transaction Fee Ethereum Virtual Machine (EVM) compatible |
| 7 | Klaytn | Permissioned | Solidity | yes | Istanbul Byzantine Fault Tolerance (IBFT) | Kakao Foundation | KLAY | High Throughput speed Reduced entry Barriers Enterprise friendly Ethereum Virtual Machine (EVM) Compatible |
| 8 | Algorand | Permissioned | Transaction Execution Approval Language (TEAL) Assembly Level Language | yes | Pure Proof of stake(PPoS) | Algorand Foundation | ALGO | Two tired Architecture Highly Scalable Low Gas Fee Highly Eco-friendly by consuming less energy |
| 9 | Fantom | supports both permissioned & Permission-less | Object oriented Languages ex. Java | yes | Lachesis | Fantom Foundation | FTM | Instant Transactions Highly Scalable Low Gas Fee Ethereum Virtual Machine (EVM) Compatible |
| 10 | Near Protocol | Permissioned | Rust, Assem-blyScript, and Solidity | yes | PoS | Near Collective | NEAR | Fast Transactions Allows transfer of tokens from other networks Low gas fee Suitable For minting NFTs |
| 11 | CORDA | Permissioned | Kotlin, Java | yes | Asynchronous Byzantine Fault Tolerance | Hedera Hashgraph Council | None | High transactional Speed Hassle-free Scalability Highly Secure Easy integration for enterprises |
| 12 | TEZOS | Permission-less | JavaScript/ TypeScript, Python, and OCaml | yes | Delegated Proof of Stake | Dynamic Ledger Solutions | Tezzie or Tez | On-chain Governance Smart contract with formal verification self-amendment protocol Functions over efficient Delegated PoS Consensus Mechanism |

### 6.21. NEO

NEO is a permissioned blockchain ledger. It used Delegated Byzantine Fault Tolerance for consensus.

NEO was designed to develop scalable decentralized applications and its main asset is the NEO token. The founders of the NEO are the duo Da Hongfei and Erik Zhang, who set up a blockchain company named OnChain, which is based in Shanghai. Delegated Byzantine Fault Tolerance is what NEO uses as its consensus algorithm and the use of NEO token is to generate GAS tokens; transaction fees can be paid by this while running applications on the network.

### 6.22. Binance Smart Chain

Binance Smart Chain (BSC) is a community-driven decentralized Blockchain network for running smart contract-based DApps. Binance Smart Chain is ideally suited for those planning to build DeFi yield

farming applications, DeFi staking projects, and synthetic asset development. Binance was initially started as a cryptocurrency exchange, but later they decided to enter into Blockchain development by launching their own Blockchain protocol. Changpeng Zhao was the brainchild behind the development of Binance, and Binance Smart Chain was launched as an alternative to the Ethereum Blockchain while having its original qualities in it. Smart Contracts can be deployed in Binance Smart Chain as they can function flawlessly being Ethereum Virtual Machine (EVM) compatible. Native tokens are valuable components of any blockchain-based system, and Binance has its native token by the name of Binance Coin (BNB). These BNB tokens are used to pay for transaction fees on the Binance Chain and Binance DEX platform, asset transfers and staking.

### 6.23. AVALANCHE

Avalanche is an open-source decentralized Blockchain platform to be developed by Ava Labs. The main aim of Avalanche is to address long-pending issues like scalability, security, and decentralization. Avalanche functions upon the Proof of Stake (PoS) mechanism that lets it maintain decentralization and security while focusing on lower costs, fast transaction speeds, and eco-friendliness. Avalanche is a very flexible platform to be written using a number of languages like Go, TypeScript, JavaScript, Python, and Vue which makes it one of the modular yet versatile protocols having its own native token by the name of AVAX. Smart contract in Avalanche are written in Solidity language that is used by Ethereum and this makes it easily interoperable. So, Avalanche can integrate a large number of DApps into DeFi that are based on the Ethereum ecosystem. Additionally, Avalanche has its own Avalanche Virtual Machine (AVM) which functions like the EVM of Ethereum, and both are compatible with each other.

Features that make Avalanche unique among other Blockchain platforms are, It has an Avalanche Virtual Machine (AVM), which can be easily integrated. It is more suitable for DeFi (Decentralized Finance) and NFT marketplaces.

### 6.24. Evaluation criteria

Each interoperability approach was evaluated on the basis of:

- **Scalability** – Ability to support multiple blockchains or high transaction volume.
- **Decentralization** – Degree to which control is distributed.
- **Security** – Mechanisms to ensure tamper-proof, verifiable operations.
- **Complexity** – Ease of integration and system design requirements.
- **Throughput**- To check successful transaction per unit time from one blockchain to another blockchain
- **Latency** -This will state the delay in time duration for successful transaction from one blockchain to another.

## 7. Discussion and future scope

As per the study of related survey interoperability of blockchain is still at infancy, many challenges are still there such as compromising decentralization in case of notary approach, 50 percent attack in case of gate way based approaches. If some one tries to solve the interoperability using sidechains or relays there is a chance of compromising the users of sidechains itself. Due to all of these things, there is a trade off within a blockchain as scalability, decentralization and interoperability of the blockchains. If user tried to get the more scalable solution for interoperability then the decentralization will get compromised. If user tried to get decentralized solution for interoperability then there are chances of compromising security of the main chain or the parallel or relay chains.

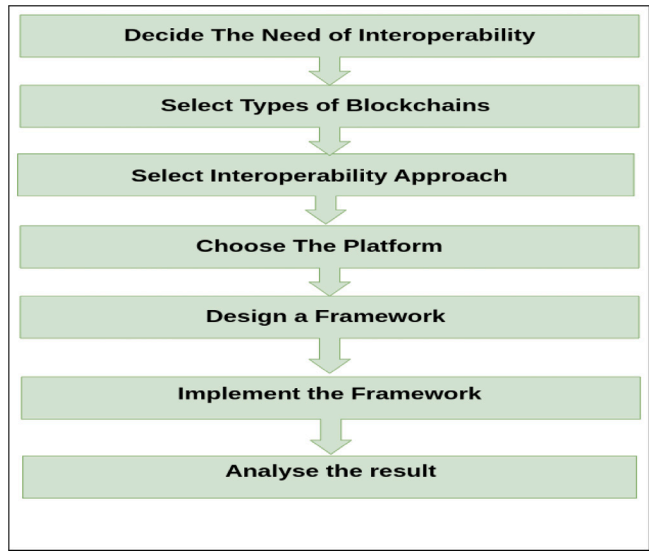


**Fig. 5.** Workflow for interoperability.

In the blockchain interoperability there are no any standards till then to prove the higher interoperability is achieved . Rather there are no any platforms who are helping to check interoperability level for blockchains. Thus the interoperability in blockchains is still depends on the use cases and its need for the interoperability. No other survey papers even mentioned about process of applicability of interoperability in blockchains. Therefore in this paper suggested a process flow is proposed to achieve correct path toward interoperability. By using that at least the interoperability strategy idea gets clear.

### 7.1. Validation of findings

To ensure the accuracy of findings, the categorization and observations were cross-validated with insights from three recent survey papers published between 2020 and 2024. In addition, a conceptual case study on crowdsourcing was mapped using the proposed workflow to demonstrate its real-world applicability.

## 8. Proposed workflow of interoperability

Looking at the literature and overall review on the state of current scenario of Interoperability in blockchains, We proposed the process flow to achieve the interoperability in blockchains. The proposed flow is as shown in Fig. 5. Proposed Workflow is a guidance for all the researchers who are working on the blockchain for cross-chain interoperability. As per the proposed process, user need to select an interoperability problem statement can be solved by the blockchains. Choice of correct use case for interoperability is not under the scope of this paper.

### 8.1. Development of the workflow

The proposed interoperability workflow was developed through synthesis of the patterns observed in the reviewed literature. While it is currently based on theoretical studies and architectural analysis, its applicability was tested conceptually on a crowdsourcing use case. Future work aims to validate the workflow using simulation or real blockchain deployment.

Using the flow as per the Fig. 5. Whenever we need to work on any use case of interoperability in blockchains first of all we need to decide the use case actually needs interoperability on which level. Also need to think of interoperability to be provided within same type of

blockchains or in between different types of blockchains, therefore need to decide type of blockchains after the use is decided. Then the next step of choice is about which type of approach is suitable for the kind of interoperability required. Once the approach is decided the particular platform can be chosen. Also as per the technologies stack requirement framework can be designed and then comes implementation and performance checking platforms to analyze the result.

## 9. Case study: Crowdsourcing using interoperability workflow

In this section, one case study is represented as an application for proposed flow by applying the workflow of interoperability to achieve blockchain interoperability.

### 9.1. Problem definition

Crowdsourcing platforms traditionally suffer from issues such as lack of trust among anonymous contributors, centralized reward systems, and limited support for cross-chain payments. To address these limitations, a blockchain-based crowdsourcing system can be designed to operate across multiple blockchains. Contributors may belong to different blockchain ecosystems and may prefer receiving rewards in various tokens. Therefore, blockchain interoperability is essential for seamless task management and cross-chain reward distribution.

### 9.2. Application of proposed workflow

The proposed interoperability workflow described in this paper is applied to the crowdsourcing use case as follows:

1. **Use Case Identification and Interoperability Need:**
   A crowdsourcing platform needs to allow contributors to submit work and receive rewards using different blockchain networks. The platform must support secure, transparent, and cross-chain communication to achieve decentralization and trustless reward distribution as either the worker receives the reward upon proof of completion, or the requester safely retrieves funds if conditions are unmet.
2. **Blockchain Type Selection:**
   The use case involves *heterogeneous blockchains* such as Ethereum (for smart contract logic), Polygon (for low-cost interactions), and Solana (for fast settlements). These blockchains differ in consensus mechanisms and token standards, necessitating heterogeneous interoperability. Crowdsourcing platforms commonly want to include many participant chains (worker wallets on different chains, requesters using other chains), so you must handle differing account models, finality, and contract capabilities — a hallmark of heterogeneous interoperability. If your paper restricts all actors to EVM-compatible chains, then homogeneous applies — but the more general and realistic crowdsourcing scenario is heterogeneous.
3. **Selection of Interoperability Approach:**
   To enable secure asset transfer and task validation across blockchains, **Hash Time-Locked Contracts (HTLC)** and **Bridge/Relay-based solutions** are suitable. HTLC ensures atomic transactions where a reward is released only if the contributor proves task completion. Relay schemes allow one blockchain to validate the state of another, ensuring data integrity.
   Making use of hash time-locked contracts (HTLC) is the one way, which implements cross-chain transactions atomically. HTLC contracts utilize time locking and pre-image revelation [16]. They enable Party A (the sender) to lock assets on the blockchain, which can later be unlocked in one of two ways: either by Party A after a set time has passed, or immediately by Party B (the receiver) if Party B can present valid proof of execution By setting two similar contracts on both blockchains, party A and party B can safely exchange their assets without a pre-negotiation or any trusted third party [16].
4. **Platform Selection:**
   Platforms such as *Polkadot*, *Cosmos*, *Atomic Loans*, and *Wormhole Bridge* are ideal choices. These platforms support interoperability through bridges, relays, and atomic swap mechanisms.
5. **Framework and Technology Stack:**
   A smart contract is deployed on Ethereum to manage task creation and proof verification. Contributors submit proofs which are hashed and verified. HTLCs are used to lock rewards on a second chain (e.g., Solana). Upon proof revelation, the reward is automatically released. Technologies include Solidity, Cosmos SDK, Chainlink CCIP, and cross-chain messaging protocols.
6. **Implementation and Analysis:**
   The implementation should be evaluated based on latency, communication overhead, transaction cost, and success rate of atomic swaps. Security concerns such as timeout fraud in HTLC and malicious relay attacks must also be analyzed.
   In the proposed workflow, security risks are considered and mitigated at each stage. During blockchain selection, differing consensus/finality models may expose the system to rollback risks; this is mitigated by waiting for sufficient confirmations. When selecting the interoperability approach, HTLCs address atomicity but may introduce timeout fraud, solved by ordering timeouts correctly and requiring collateral from participants. In framework design and implementation, risks such as relay attacks, replay attacks, and oracle manipulation are managed by decentralized relayer sets, fraud-proof mechanisms, nonces, and multi-source oracles. Finally, during analysis of results, simulation and testnet experiments are used to test adversarial conditions and fine-tune security parameters.

### 9.3. Benefits and observations

Applying the proposed interoperability workflow to the crowdsourcing use case provides the following benefits:

- **Decentralized reward distribution** without intermediaries.
- **Multi-chain participation**, increasing contributor diversity and platform reach.
- **Trustless task verification and execution** via smart contracts and cross-chain proof mechanisms.
- **Flexible reward models** in different tokens as per contributor preference.
- **Simulations and testnets** can be utilized for the usecase implementation using proposed interoperability flow as follows:
  cross-chain message delays, malicious relayers, timeout, workload (many tasks and claim attempts), attacker models Recommended validation steps: unit and integration tests for smart contract on each cain (simulate HTLC flows, refunds, edge cases)
  Local multi-node testnets (e.g., ganache/hardhat instances + a Fabric localnet or Substrate dev nodes) to reproduce heterogeneous environments.
  Interoperability test harness/discrete-event simulator to model message propagation delays, timeouts, and malicious actors deterministically.
  Public testnets or private consortium testnets to evaluate network effects, gas costs, and real-world latencies.
  Benefits: experimental validation demonstrates liveness under realistic conditions, quantifies timeout parameters and cost tradeoffs, exposes security gaps, and provides reproducible benchmarks — all of which strengthen claims and guide parameter choices in the paper.

This case study validates the relevance and applicability of this proposed interoperability framework in practical, real-time scenarios like crowdsourcing. There are a few more use cases such as healthcare as well as supply chain. One of the use case of healthcare's interoperability framework can be designed as per the paper by us as [6] using the proposed workflow.

## 10. Limitations of the methodology

Although this survey offers a comprehensive overview of blockchain interoperability approaches and proposes a structured workflow for implementation, certain limitations in the methodology are acknowledged. These do not detract from the overall contribution of the work but rather provide direction for future research improvements.

1. **Lack of Experimental Validation:**
   This work is based on literature review and conceptual analysis. No empirical evaluation, such as simulation or performance benchmarking of interoperability models, has been conducted. Future work can incorporate experimental validation to support practical applicability.
2. **Absence of Standardized Evaluation Metrics:**
   The comparison among interoperability approaches is qualitative due to the absence of universally accepted metrics for measuring interoperability effectiveness. Establishing standard benchmarks in future studies would enhance comparative analysis.
3. **Generic Nature of the Proposed Workflow:**
   The proposed interoperability workflow is designed as a generalized model. However, its adaptability to domain-specific applications (e.g., healthcare, finance, supply chain) remains to be validated. Domain-focused refinements can improve its applicability.
4. **Platform Evolution and Versioning Constraints:**
   Blockchain platforms evolve rapidly with frequent updates. The methodology captures the current state at the time of writing this paper; however, periodic revisions will be essential to ensure alignment with technological advancements.

These limitations highlight meaningful opportunities for future research and development in the field of blockchain interoperability. Addressing them can significantly strengthen the methodological robustness and practical utility of such studies.

## 11. Conclusion and future scope

This research was conducted a thorough assessment of the literature on blockchain interoperability for this paper. This research thoroughly examined, contrasted, and talked about 35 documents, which corresponded to blockchain interoperability options. It hope to circumvent inherent constraints on the blockchain interoperability research field, This study offers intriguing findings that fall into two categories: Approaches, Platforms. The systematic exploration of each solution is what makes the study unique.

Blockchain interoperability is not limited to this research. But as this research proposed a flow for solving use cases of interoperability in blockchains. This will help researchers or developers to work under the interoperability of blockchain in a systematic way. Thus this research will help to reach out correct level of interoperability for the blockchains and therefore blockchain interoperability will grow gradually with respect to this perspectives.

In future, security and privacy can be enhanced with different secured methodologies such as ZKP (Zero Knowledge Proofs) [48], different encryption techniques etc. Also the paper's proposed workflow can be utilized to apply on usecases such as healthcare, supply chain where, multiple blockchains are involved.

## CRediT authorship contribution statement

**Rohini V. Tate:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Sunil B. Mane:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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